Aditya Kulkarni

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I develop theoretical and computational frameworks for problems spanning high energy physics, differential privacy, and modern data driven methods. My research combines analytic modeling, statistical inference, and machine learning techniques to study complex physical and computational systems.

EDUCATION

University of Massachusetts Amherst **Aug. 2020 – May 2027 (Expected)**
Doctor of Philosophy, Physics *Amherst, MA*
- Advisor: Prof. Lorenzo Sorbo.

State University of New York, Stony Brook **Aug. 2018 – Jul. 2020**
Master of Arts, Physics *Stony Brook, NY*
- Advisor: Prof. Abhay Deshpande.

Savitribai Phule Pune University **Jul. 2014 – Jun. 2018**
Bachelor of Engineering, Mechanical *Pune, India*
- Advisor: Prof. S. D. Limaye.

PUBLICATIONS

- Kulkarni et al. "How Low Can You Go: Fine-Tuning LLMs with Bayesian Optimization (LlMBO)", Oct. 2026. arXiv:2610.18504 [stat.ML]. Preprint.

- Aditya Kulkarni and Carlos Soto. "Differentially private geodesic regression". In Proceedings of the 43rd International Conference on Machine Learning, Proceedings of Machine Learning Research. PMLR, 2026.

- A. Kulkarni and L. Sorbo, "The onset of the strong backreaction regime in axion inflation, an analytical study," Aug. 2026. arXiv:2608.29244 [astro-ph.CO]. Preprint.

- A. Kulkarni and L. Sorbo, "Spectra of fermions produced by a time-dependent axion in the radiation- and matter-dominated universe," Journal of Cosmology and Astroparticle Physics, vol. 2025, no. 01, p. 039, Jan. 2025. doi: 10.1088/1475-7516/2025/01/039. [Online]. Available: `https://dx.doi.org/10.1088/1475-7516/2025/01/039`.

In progress:

- Aditya Kulkarni, et al. "Differential Privacy over Lie Groups."

INVITED TALKS AND PRESENTATIONS

ICML (International Conference on Machine Learning) **July 6–11, 2026**
Differentially Private Geodesic Regression *Seoul, South Korea*

PPPC (Privacy and Public Policy Conference) **Feb. 9–10, 2026**
Differentially Private Geodesic Regression *Washington DC, USA*

Conference on Particle Physics & Cosmology **August 23–27, 2026**
Analytical study of strong backreaction *Leiden, Netherlands*

School Of Basic Sciences **August 31–Sept. 4, 2026**
Analytical study of strong backreaction *Campania, Italy*

TEACHING HISTORY

Instructor **Spring 2023**
University of Massachusetts, Amherst *Amherst, MA*
- Full instructor for PHY 151: Introduction to Newtonian mechanics; designed lectures, homework, examinations, and grading policies, and coordinated teaching assistants for more than 200 students.

Teaching Assistant **Fall 2026**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 132 labs: Focus on data collection and statistical analysis.

Teaching Assistant **Spring 2026**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 132 labs: Focus on data collection and statistical analysis.

Teaching Assistant **Fall 2025**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 132 labs: Focus on data collection and statistical analysis.

Teaching Assistant **Spring 2025**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 132 labs: Focus on data collection and statistical analysis.

Teaching Assistant **Fall 2024**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 151 labs: Engineering mechanics with focus on data collection and statistical analysis.

Teaching Assistant **Spring 2024**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 151 labs: Engineering mechanics with focus on data collection and statistical analysis.

Teaching Assistant **Fall 2023**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 440 labs: Senior labs with focus on data collection and statistical analysis.

Teaching Assistant **Fall 2022**
University of Massachusetts, Amherst *Amherst, MA*
- Grader for PHY 615: Grading Quantum mechanics II.

Teaching Assistant **Spring 2022**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 131 labs: Engineering mechanics with focus on data collection and statistical analysis.

Teaching Assistant **Fall 2021**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 440 labs: Senior labs with focus on data collection and statistical analysis.

Teaching Assistant **Spring 2021**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 132 labs: Focus on data collection and statistical analysis.

Teaching Assistant **Fall 2020**
University of Massachusetts, Amherst *Amherst, MA*
- Teaching Assistant for PHY 181 labs: Focus on data collection and statistical analysis.

STATISTICS AND MACHINE LEARNING RESEARCH

Bayesian Optimization for LLM Fine-Tuning **August 2026 – Present**
Advisor: Prof. Carlos Soto, Prof. Nathan Wycoff *University of Massachusetts, Amherst*

- Developed an inference only Bayesian optimization framework for parameter efficient language model fine-tuning, connecting Gaussian process trust-region optimization with low dimensional TinyLoRA parameterizations.
- Designed comparisons with policy-gradient training across model families and reasoning benchmarks, using shared training data, identical projection seeds, and matched generated completion budgets.

Differential Private Machine Learning **May 2024 – June 2025**
Advisor: Prof. Carlos Soto *University of Massachusetts, Amherst*

- Developed intrinsic differentially private geodesic regression, deriving sensitivity bounds through Jacobi field analysis and connecting privacy calibration to manifold geometry; published at ICML 2026.
- Implemented the methods and computational experiments on spheres, symmetric positive definite matrices, and Kendall's planar shape space; evaluated regression error across sample sizes and privacy-budget allocations.

Domain Adaptation **Aug. 2025**
University of Massachusetts Amherst *Amherst, MA*

- Developed a domain adaptation pipeline for astronomical imaging, transferring models trained on simulated galaxy images to real survey data to improve galaxy morphology classification and gravitational-lens Einstein radius estimation. Leveraged optimal transport and Sinkhorn alignment methods to reduce sim→real domain shifts caused by differing noise, PSF, and redshift distributions.
- Benchmarked multiple adaptation strategies (energy-based, optimal transport, Sinkhorn) against non adapted baselines, demonstrating improved classification accuracy and radius predictions, supporting more reliable identification of strong lensing events in large-scale surveys.

PHYSICS RESEARCH EXPERIENCE

Axion Evolution from Weak to Strong Backreaction. **September 2024 – July 2026**
Advisor: Prof. Lorenzo Sorbo *University of Massachusetts, Amherst*

- Developed a semi-analytical framework for homogeneous axion inflation with Chern–Simons gauge-field production, describing the transition from weak to strong gauge field backreaction on the axion background.
- Studied the onset of inhomogeneous backreaction semi analytically, identifying when it arises and how its behavior depends on key model parameters such as the axion–gauge coupling and background dynamics.

Fermion Production by Time-Dependent Axion **May 2023 – Aug. 2024**
Advisor: Prof. Lorenzo Sorbo *University of Massachusetts, Amherst*

- Developed a fully non-perturbative Bogolyubov-based framework to analyze axion–fermion particle production in a spatially flat FLRW universe.
- Showed how rolling and oscillating axion backgrounds populate fermionic states across broad momentum scales during radiation and matter eras.
- Identified helicity asymmetric, non-relativistic production as a potential matter like energy source influencing the cosmic energy budget, CMB polarization, and large scale structure.

Moller Experiment: Photon Shielding **Aug. 2020 – Aug. 2021**
Advisor: Prof. Krishna Kumar *University of Massachusetts, Amherst*

- Designed and optimized photon shielding for high energy backgrounds in the Møller integrating Cherenkov detectors using GEANT4 Monte Carlo simulations which modeled photon interactions, scattering, and background suppression for precise A_{pv} measurement within the segmented ring detectors and tight geometric constraints.
- Engineered critical structural modifications to the Møller detector assembly using SOLIDWORKS CAD, ensuring mechanical alignment and compatibility of shielding with the toroidal spectrometer, collimator, and Cherenkov detector geometry while respecting experimental access and stability requirements.

SoLID Mirror Production & Testing (Master's Thesis) **Aug. 2019 – Jul. 2020**
Advisor: Prof. Abhay Deshpande *State University of New York, Stony Brook*

- Designed and executed the in-house fabrication of precision spherical mirrors for SoLID light- and heavy-gas Cherenkov detectors, optimizing physical vapor deposition parameters (substrate preparation, Al/Cr adhesion layers, MgF_2 coatings, and vacuum conditions) to maximize ultraviolet reflectivity and ensure accurate focusing of Cherenkov photons onto photomultiplier arrays.
- Analyzed thin-film thickness uniformity and reflection efficiency using geometric modeling and thin-film interference principles, validating optical performance against detector requirements for SIDIS and PVDIS particle identification.

TECHNICAL SKILLS

Languages: Python, R, SQL, Mathematica
Libraries/Frameworks: Scikit-learn, statsmodels, TensorFlow, Linux, GitHub, Geant4, ROOT

MEMBERSHIP

* American Statistical Association (ASA).
* American Mathematical Society (AMS).
* American Physical Society (APS).